Leggett-Garg Inequality-based Quantum Computers Quality Benchmark

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Motivation

- Need for independent benchmarking tools
- With accordance to European Quantum Strategy
- The use of Star-topology devices in the advent of Constellation-type devices
- ...

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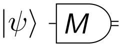
Executive summary (judging panel made up of Department Heads, Management, and VPs of Technology)

- What we did
- Additional technical results (Star Device Qubit Selector, FakeSirius)

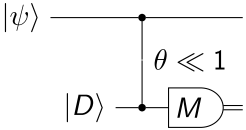
Projective vs weak measurement

Projective (strong)



- Direct
- Accurate (in theory)
- Disturbing (decoherence)

Weak



- Indirect (weak coupling)
- Inaccurate (deliberately)
- small decoherence

No free lunch: information gain $> \theta$, decoherence $> \theta^2$.

Application: Leggett-Garg inequality violation

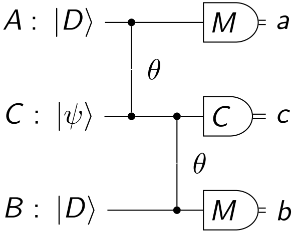
$$\langle a\rangle + \langle b\rangle - \langle ab\rangle \le 1,\ |a| \le 1, |b| \le 1$$

The catch: strength calibration θ: $a_{raw} = \theta a_{physical}$

- Strength-agnostic form
- postselection $c = 0, 1$,
- $A \equiv B$ (identical, independent)

$$Q = \frac{\langle a + b|c\rangle^2}{4\langle ab|c\rangle} \le 1$$

$\langle ab|c\rangle = \langle abc\rangle / p(c = 1)$

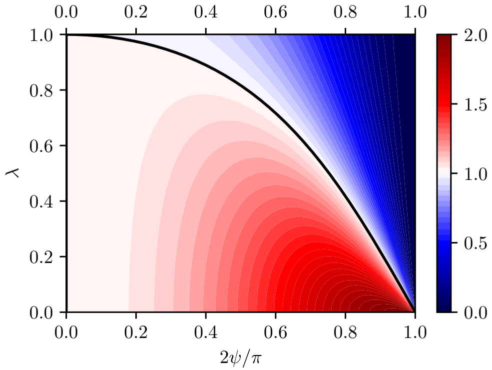


Objective Reality: Description + formulas from LGA3

A test with strength dependence on $\lambda = \sin\theta$

$$A = B = Z = |0\rangle\langle 0| - |1\rangle\langle 1|,\ C = |\psi_-\rangle\langle\psi_-|,\ |\psi\rangle = |\psi_+\rangle$$

$$|\psi_\pm\rangle = \cos\frac{\psi}{2}|1\rangle \pm \sin\frac{\psi}{2}|0\rangle$$

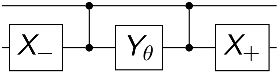


Weak coupling implementation

- two qubit gate $U \sim I$, basis $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$
- IBM/IonQ – native

$$RZZ(\theta) = \mathrm{diag}(1, e^{i\theta}, e^{i\theta}, 1)$$

- IQM – two CZ sandwiching Y_θ

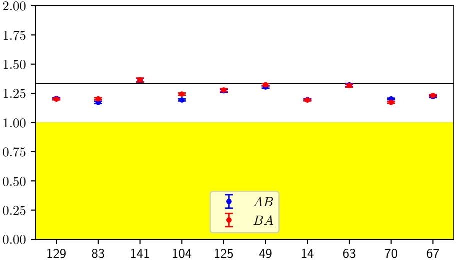


Notation: $X = |0\rangle\langle 1| + |1\rangle\langle 0|$, $Y = i|1\rangle\langle 0 - i|0\rangle\langle 1|$, $V_\theta = \exp(-i\theta V)$, $V_\pm = V_{\pm\pi/2}$

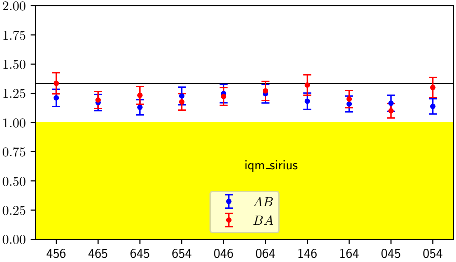
Objective Reality Results

Results on IBM/IonQ/IQM $\theta = 0.1$

IBM kingston (C qubit specified)
$2 \cdot 10^7$ shots



IQM Sirius (CAB qubits specified)
10^6 shots



IonQ Forte 1: 1.272 ± 0.035 ($A \to B$), 1.289 ± 0.035 ($B \to A$)

Proposed benchmark results?

IQM Pulla failed results

Impact and roadmap

Challenges

- more access needed to lower error $\sigma \sim 1/\theta^2\sqrt{N}$,
- pulse control $U \to U^\theta$, e.g. $CZ \to \mathrm{diag}(1,1,1,e^{\pm i\theta})$, continuous families of entangling gates
- avoid postselection (full noninvasiveness)
- Star topology $\to$ scale up the test, more detectors

Benefits

- benchmarking partially entangling gates and pulse control
- proposed benchmarking score/map (for the control range)

$$Q-1,\ (Q-1)/\sigma$$

- minimal disturbance: $\theta = 1/10$ disturbance $1/400$.
- robustness to noise – no error correction needed

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Scalability

Thank you for your attention!

https://github.com/Tomev/IQM_Research_Award_2026

Last but not least!

- our long-standing collaborators/supporters: Wolfgang Belzig, Jakub Tworzydło, and Josep Batle
- to read more https://arxiv.org/abs/2603.22020
- IBM Quantum, IonQ, IQM – access provided
- Poznań Supercomputing and Networking Center – Polish access hub
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